

Human:

Your job is to help me fully understand the MuP (Maximal Update Parametrization) topic by creating a couple of practice problems with varying difficulty. I have attached lecture notes and the MuP homework questions as well as the MuP research paper.

You should create 1 conceptual problem and 2 problems that closely represent the homework's difficulty and style.

You should prompt each question one at a time and if I solve it correctly, move on. If I am incorrect, please say so and provide hints towards the correct solution.

Claude:

I'll help you master MuP by creating practice problems that build on the concepts from your lecture notes and homework. Let me start with one problem at a time.

Problem 1: Conceptual Understanding

Consider two different approaches to training neural networks:

I really like this question as it is simple but really tests the core idea.
Because this content is fairly old, it was really useful in refreshing my memory.

Approach A (Standard Parameterization):

- Initialize weights W with Xavier scaling: $W \sim N(0, 1/d_{in})$
- Use Adam with learning rate η that doesn't scale with layer width

Approach B (Maximal Update Parameterization):

- Initialize weights W with Xavier scaling: $W \sim N(0, 1/d_{in})$
- Use Adam with learning rate η that scales as $\eta \propto 1/d_{in}$

Question: Suppose you train a small model (width 128) and find the optimal learning rate is $\eta = 0.001$. You now want to train a wider model (width 1024) with the same architecture.

(a) Under Approach A, what would happen if you use the same learning rate $\eta = 0.001$? Would you expect it to be too large, too small, or about right?

(b) Under Approach B, what learning rate should you use for the wider model, and why would this allow "zero-shot hyperparameter transfer"?

Take your time and explain your reasoning!

Human:

a) The learning rate will be too big. This will result in overshooting the training which could result in no convergence.

b) We should use a learning rate that scales with $1/1024$. Since we find 0.001 to be the optimal, we solve for the pre scale learning rate: $0.001 = (1/128) * x$. $x = 0.128$. Thus, the learning rate should be $0.128 / 1024$.

Claude:

Excellent work! Your answers are **correct** for both parts. Let me break down why:

(a) Correct! Under Standard Parameterization, the learning rate of 0.001 would indeed be **too large** for the wider network. Here's the key insight from the homework:

- When using SignGD/Adam without width-dependent scaling, the RMS norm of feature updates scales as $\Theta(d_{in})$
- So going from width 128 $\rightarrow$ 1024 means updates become $\sim 8\times$ larger
- This causes overshooting and training instability, exactly as you said

(b) Correct! Your calculation is spot-on:

- At width 128: $\eta = 0.001 = (1/128) \times c$, so $c = 0.128$
- At width 1024: $\eta = 0.128/1024 = \mathbf{0.000125}$

This demonstrates the "zero-shot hyperparameter transfer" - you can tune on a small model and mathematically transfer the optimal learning rate to any width!

Great job! Let's move to a more computational problem.

Problem 2: Feature Norm Analysis (Medium Difficulty)

Consider a 2-layer MLP with MuP scaling:

- Layer 1: $W_1 \in \mathbb{R}^{(512 \times 256)}$, initialized with Xavier scaling
- Layer 2: $W_2 \in \mathbb{R}^{(128 \times 512)}$, initialized with Xavier scaling
- Input: $x \in \mathbb{R}^{256}$ with entries sampled i.i.d. from $N(0,1)$

The forward pass computes: $h_1 = W_1 x$, then $y = W_2 h_1$ (no nonlinearity for simplicity).

Question: If we want $\|h_1\|_{\text{RMS}} = \Theta(1)$, what constant should we multiply W_1 by? Then, given this scaling, what is the expected value of $\|y\|^2$?

Show your work!

The did not like how this question is given with exact numbers.
I think using variables would be nicer since it shows more of the patterns. However, I think this was a great question since it gets into the details of the scaling in MuP.

Human:

$$\|h_1\|_2 = \|W_1\|_2 * \|x\|_2$$

$$\|h_1\|_{rms} = 1/\sqrt{d_{out}} * \|h_1\|_2 \text{ which implies we want } \|h_1\|_2 = O(\sqrt{d_{out}}).$$

Since x is $N(0, 1)$, $\|x\|_2 = O(\sqrt{d_{in}})$.

Thus, we want $\|W_1\|_2 = O(\sqrt{d_{out} / d_{in}})$. However, Xavier means that each row in W_1 has variance $1/256$. This means it satisfies this condition already, so thus we scale by 1.

The expected value of each squared component in y is 1 because of how y is W_2 and H_1 is made. Thus, summing them would mean the expected value is 128.

Claude:

Excellent work! Your reasoning is correct on both parts. Let me confirm:

(Part 1: Scaling W_1) ✓ Correct! With Xavier initialization, W_1 already satisfies the desired property:

- Each entry $W_{1ij} \sim N(0, 1/256)$
- For component $(h_1)_i$: $E[(h_1)_i^2] = 256 \times (1/256) \times 1 = 1$
- So $E[\|h_1\|_2^2_{RMS}] = 1 \implies \|h_1\|_{RMS} = \Theta(1)$ ✓
- **Scaling factor: 1** (no additional scaling needed)

Additionally, it is nice how it gives a small recap on the reason why this is an important problem and what to takeaway (key insight)

(Part 2: Expected value of $\|y\|_2^2$) ✓ Correct!

- $W_2 \in \mathbb{R}^{128 \times 512}$ with Xavier initialization (variance $1/512$)
- Each h_1 component has variance ≈ 1 (from part 1)
- Each y_i component: $E[y_i^2] = 512 \times (1/512) \times 1 = 1$
- Total: $E[\|y\|_2^2] = 128 \times 1 = 128$ ✓

Great job! You're understanding the key insight: **Xavier initialization already implements MuP scaling for the forward pass**, keeping RMS norms at $\Theta(1)$ across layers.

Now let's tackle the final problem, which combines initialization and training dynamics:

Problem 3: MuP Training Dynamics (Harder)

In this question, I like how Claude pushed me to provide a more detailed answer.

Consider a single dense layer with a nonlinearity:

- Input: $x \in \mathbb{R}^{d_{in}}$ with $\|x\|_{RMS} = 1$
- Weights: $W \in \mathbb{R}^{d_{out} \times d_{in}}$, initialized with Xavier scaling
- Forward pass: $h = \sigma(Wx)$ where σ is applied element-wise

You train with SignGD (simplified Adam) for one step on a single example with loss ℓ . The gradient update is:

$$W_{t+1} = W_t - \eta \cdot \text{sign}(g \cdot x^T)$$

where $g \in \mathbb{R}^{d_{out}}$ is the backpropagated gradient vector.

Question: Assume MuP scaling where $\eta = c/d_{in}$ for some constant c .

(a) Show that $\|\Delta W \cdot x\|_{RMS} = O(1)$ regardless of d_{in} and d_{out} .

(b) During training, both W and x are changing. If we want $\|\Delta h\|_{RMS} = O(1)$ where $\Delta h = \Delta W \cdot x + W \cdot \Delta x$, what additional condition do we need on $\|\Delta x\|_{RMS}$?

Show your reasoning step by step!

Human:

a) For a single entry j in Δh , we get $\Delta W_{ij} * x_i$. x_i is RMS $O(1)$ and ΔW_{ij} is RMS $O(1)$. Thus, the RMS norm is $O(1)$.

b) Since we assume ΔW is rms norm = 1, and given x is rms norm 1, the first term is $O(1)$. Since W is Xavier scaled, the RMS norm is 1. Thus we need the condition that Δx is rms 1.

Claude:

Good intuition, but let me ask you to be more precise on (a) - your reasoning is too hand-wavy!

Part (a) needs more detail:

You said " ΔW_{ij} is RMS $O(1)$ " but let's actually calculate this. Remember from the homework setup:

- $\Delta W = -\eta \cdot \text{sign}(g \cdot x^T)$ where this is an outer product
- So each entry $(\Delta W)_{ij} = -\eta \cdot \text{sign}(g_i) \cdot \text{sign}(x_j)$

Claude tries to push me to explain why an Xavier scaled weight has an RMS norm of 1. However, in earlier parts I made this assumption without proving it as well and Claude did not try to make me explain further. I also made the assumption in part b of this question.

Now consider a **single output component**: $(\Delta h)_i = \sum_j (\Delta W)_{ij} \cdot x_j = -\eta \cdot \text{sign}(g_i) \cdot \sum_j \text{sign}(x_j) \cdot x_j = -\eta \cdot \text{sign}(g_i) \cdot \sum_j |x_j|$

Can you complete the calculation from here?

What is $E[(\Delta h)_i^2]$? (Remember each $|x_j|$ has expected value $\Theta(1)$, and there are d_{in} terms)

Then, what is $\|\Delta h\|^2_{\text{RMS}} = \|\Delta h\|^2_2 / d_{\text{out}}$?

Finally, substitute $\eta = c/d_{\text{in}}$ and show you get $O(1)$.

Part (b) is correct! ✓ Your logic is right: if $\|W\|_{\text{RMS} \rightarrow \text{RMS}} = O(1)$ (from Xavier) and $\|\Delta W \cdot x\|_{\text{RMS}} = O(1)$ (from part a), then to keep $\|\Delta h\|_{\text{RMS}} = O(1)$, we need $\|\Delta x\|_{\text{RMS}} = O(1)$.

Try finishing part (a) with the full calculation!

Human:

I don't really understand why the norm is 1 for part a. Please explain it to me.

Claude:

Great question! Let me walk through the calculation carefully, step by step. This directly follows homework problem 1(b).

Step-by-Step Calculation:

We have the update: $\Delta W = -\eta \cdot \text{sign}(g \cdot x^T)$

So each entry is: $(\Delta W)_{ij} = -\eta \cdot \text{sign}(g_i) \cdot \text{sign}(x_j)$

For a single output component:

$$\begin{aligned} (\Delta h)_i &= \sum_j (\Delta W)_{ij} \cdot x_j \\ &= -\eta \cdot \text{sign}(g_i) \cdot \sum_j \text{sign}(x_j) \cdot x_j \\ &= -\eta \cdot \text{sign}(g_i) \cdot \sum_j |x_j| \end{aligned}$$

Now let's find $E[(\Delta h)_i^2]$:

Since $\text{sign}(g_i)^2 = 1$:

$$E[(\Delta h)_i^2] = \eta^2 \cdot E[(\sum_j |x_j|)^2]$$

Key insight: Each $x_j \sim N(0,1)$, so $E[|x_j|] = \sqrt{2/\pi} \approx 0.798 = \Theta(1)$

Therefore:

- $E[\sum_j |x_j|] = d_{\text{in}} \cdot E[|x_j|] = \Theta(d_{\text{in}})$
- $E[(\sum_j |x_j|)^2] = \Theta(d_{\text{in}}^2)$

I like the detailed explanation it made for this followup. I also liked how it gives the homework question to reference.

So:

$$E[(\Delta h)_i^2] = \eta^2 \cdot \Theta(d_{\text{in}}^2)$$

Now substitute $\eta = c/d_{\text{in}}$ (MuP scaling!):

$$\begin{aligned} E[(\Delta h)_i^2] &= (c/d_{\text{in}})^2 \cdot \Theta(d_{\text{in}}^2) \\ &= c^2/d_{\text{in}}^2 \cdot \Theta(d_{\text{in}}^2) \\ &= c^2 \cdot \Theta(1) \end{aligned}$$

Finally, compute the RMS norm:

$$\begin{aligned} \|\Delta h\|^2_{\text{RMS}} &= (1/d_{\text{out}}) \cdot \sum_i E[(\Delta h)_i^2] \\ &= (1/d_{\text{out}}) \cdot d_{\text{out}} \cdot c^2 \cdot \Theta(1) \\ &= \Theta(1) \quad \checkmark \end{aligned}$$

The key insight: Without the $1/d_{\text{in}}$ scaling (i.e., if $\eta = c$ constant), you'd get:

- $E[(\Delta h)^2] = c^2 \cdot \Theta(d_{in}^2)$ — **blows up with width!**

But with MuP scaling $\eta = c/d_{in}$:

- The $1/d_{in}^2$ from η cancels the d_{in}^2 from the sum
- Result: $\|\Delta h\|_{RMS} = \Theta(1)$ **regardless of width!**

Does this make sense now?
